FEDERAL SPECIFICATION

BOTTLE, STERILE FLUID STORAGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of square-shaped glass bottle, fitted with an automatic-sealing closure cap, suitable for the preparation and storage of irrigating solutions under vacuum for prolonged periods (see 6.1).

2. APPLICABLE SPECIFICATIONS, AND STANDARDS

2.1 **Specifications and standards.**—The following specifications, standards, and drawings of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L–P–406 — Plastic, Organic; General Specifications, Test Method.

PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 601—Rubber; Sampling and Testing.

Fed. Test Method Std. No. 626—Glassware; Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—Bottle shall be made of heat-resistant, transparent, lead-free, borosilicate

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glass, having a maximum resistance to the corrosive action of distilled water. The glass shall have a linear coefficient of expansion of 32 x 10^{-7} to 41 x 10^{-7} per degree centigrade (C.) between 0° and 350° C., when tested as specified in 4.4.1.1. The glass shall not require more than 1.0 milliliter (ml.) of 0.02 normal sulfuric acid to neutralize alkalinity per 10 grams of glass, when tested as specified in 4.4.1.1. The glass shall be fully annealed, and shall show no ring strain when tested as specified in 4.4.2.1.1. Any longitudinal strain, if present, shall be faint and highly diffused. The bottles shall be capable of withstanding 45 pounds per square inch of bursting pressure when tested as specified in 4.4.2.1.2.

3.1.2 *Plastic.* — The closure cap shall be made of a black phenolic-type plastic molding material, having the following physical characteristics, when tested as specified in 4.4.1.2.

Water absorption, percent	0.4 maximum
Distortion under heat—	
°Fahrenheit (F.)	320 minimum

There shall be no appreciable changes in the above physical properties after the plastic specimens have been subjected to the deterioration tests specified in 4.4.1.2.1. The finished cap shall withstand autoclaving, as specified in 4.4.2.2, without distortion.

3.1.3 *Rubber.*—The collar shall be made of a black, nontoxic, nonblooming, neoprene rubber material, having the following characteristics, when tested as specified in 4.4.2.3.

Durometer hardness (unaged)	48 maximum
Durometer hardness (after autoclave aging)	53 maximum

3.2 **Style and design.**—Sterile fluid storage bottle shall consist of a bottle, cap, and collar in accordance with figures 1, 2, and 3 respectively.

3.2.1 *Bottle.*—The bottle shall be molded of material as specified in 3.1.1, and shall have an average thickness of 3/16-inch with no section having a thickness of less than 3/32-inch. The mouth of the bottle shall be molded as shown on figure 1. In order to allow for evaporation during sterilization, the filling mark shall be at a volume of 1050 cubic centimeters (cc.) ± 2 percent, when tested as specified in 4.4.2.1.3. All four sides of the bottle shall have molded in raised letters, 1/4 inch minimum in size, "For External Use Only", as shown on figure 1. On two opposite sides, the phrase "Capacity—1000 cc." shall be molded in raised letters, 3/16 inch minimum in size, near the bottom of the bottle. On the sides upon which the capacity is marked and at a mark corresponding to the 1050-cc. level, shall be a raised line, 1/16 inch wide and no less than 1½ inches long, which shall represent the solution level when filled. Under this line, in molded raised letters, 3/16 inch minimum in size, shall be the phrase "Fill to Here".

3.2.2 *Closure cap.*—The closure cap shall be constructed in accordance with figure 2. The cap shall automatically seal itself to the rubber collar so as to retain a vacuum of 29½ inches of mercury after sterilization, as specified in 4.4.1.2.1.1, when the bottle is filled to the mark with solution.

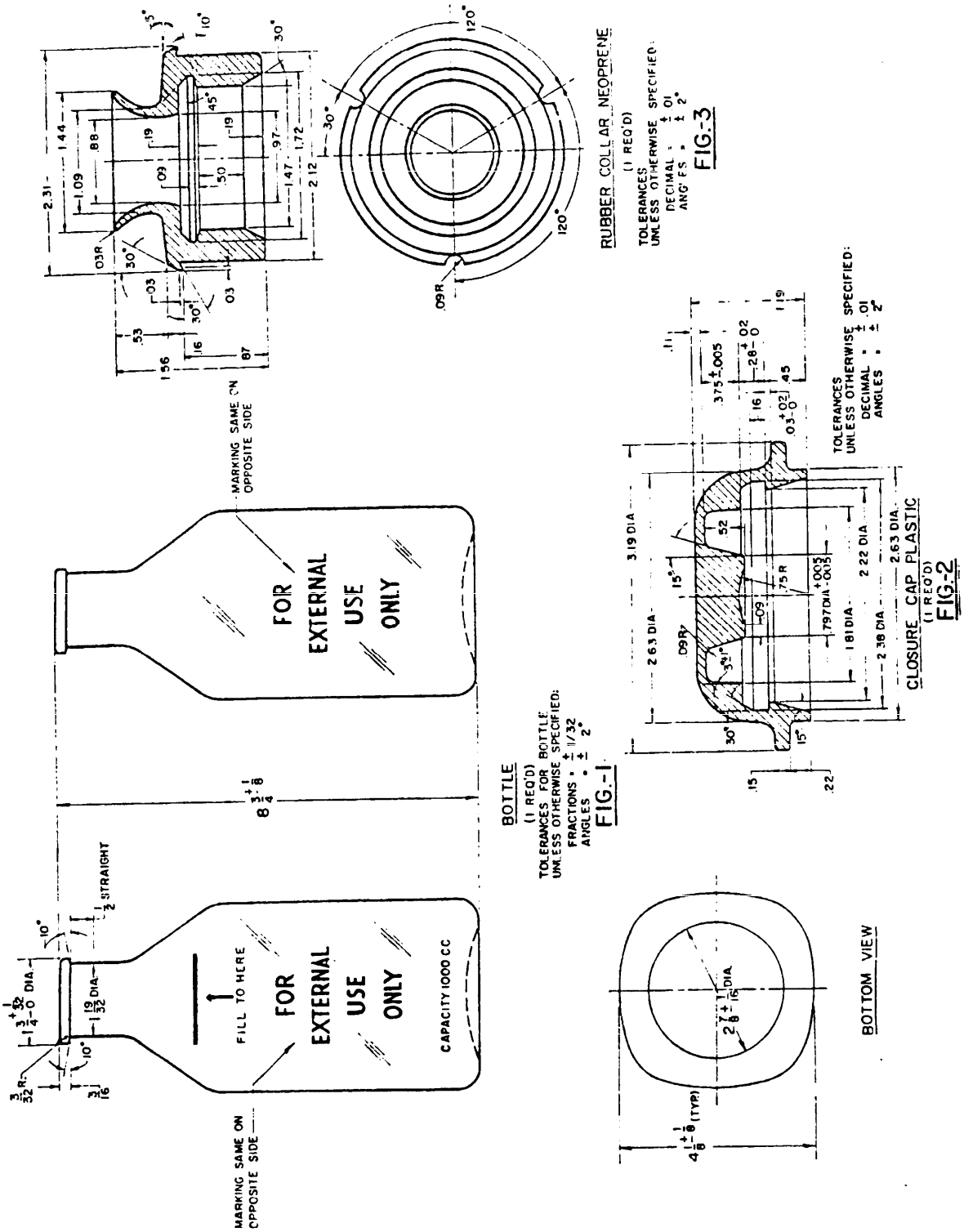
3.2.3 *Collar.*—The automatic closure collar shall be in accordance with figure 3. The inside of the collar shall have a groove, shaped as shown on figure 3, to grip the beaded neck of the bottle in order to retain the vacuum specified in 3.2.2.

3.3 **Finish.**

3.3.1 *Bottle.*—Glass shall be properly distributed and well annealed. Glass shall be practically free of defects such as rough spots, blowovers, bubbles, and blisters.

3.3.2 *Collar and cap.*—All edges shall be free from sharp, burred, and ragged edges. The cap and collar shall be uniformly colored. The cap and collar shall be free of matting,

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cracks, bubbles, and pitting. Collar and cap shall be symmetrically formed.

3.4 **Identification marking.**—The name or registered trademark of the manufacturer shall be molded into the collar in raised letters of appropriate size, and shall appear on the top of the plastic cap.

3.5 **Workmanship.**—Workmanship shall be first class throughout. The sterile fluid storage bottles shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection** —Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**—Sampling shall be conducted in accordance with Military Standard MIL–STD–105. The unit of product for inspection purposes shall be the individual bottle.

	Inspection level	*AQL (percent defective)*
For visual examination:		
Major defects	II	1.0
Total defects (major and minor defects combined)	II	4.0
For dimensional examination	L5	2.5

For materials testing.—Samples of the glass and plastic, prepared in conformance with test methods listed in 4.4.1, shall be selected for test. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated for inspection level L2 in Military Standard MIL–STD–105. No test result shall fall below the minimum or exceed the maximum values specified. Not less than 2 samples shall be taken, regardless of lot size.

For end item testing:

Component (unit of product)	Characteristic	Req	Test procedure	Inspection level	AQL
Bottle	Annealing	3.1.1	4.4.2.1.1	L4	1.0
	Bursting pressure	3.1.1	4.4.2.1.2	L4	1.0
	Vacuum retention	3.2.2	4.4.1.2.1.1	L4	1.0
	Capacity	3.2.1	4.4.2.1.3	L2	1.0
Cap	Autoclaving	3.1.2	4.4.2.2	L4	1.0
Collar	Hardness	3.1.3	4.4.2.3	L4	1.0

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4.2.1 When possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.**—The bottles shall be examined to determine compliance with all requirements contained in this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.1 *Classification of defects.*—Examination shall be conducted in accordance with the following classification of defects:

*Categories and defects**

Major.

101. Bottle and cap not free from cracks, breaks, chips, sharp edges, and deformed areas.
102. Collar not free from cuts, tears, ruptures, and holes.
103. Bottle not stable; rocks on flat surface.
104. Collar fails to fit readily into plastic cap.
105. Cap fails to seat and unseat properly over collar mounted on the mouth of bottle.

Minor.

201. Bottle not substantially free from seeds, striae, and bubbles.
202. Raised marking on bottle not present or complete.
203. Marking on cap not present and legible.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 *Dimensional examination.*—The bottles shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 **Tests.**—Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Materials.*

4.4.1.1 *Glass.*—Glass used in the construction of the bottles shall be tested for the physical properties listed, in accordance with test methods of Federal Test Method Standard No. 626, as follows:

Chemical durability ____method 320 (method P-W)
Coefficient of expansion_method 240 or 241

4.4.1.2 *Plastic.*—The plastic molding material used in the construction of the cap shall be tested for the properties listed, in accordance with the test methods of Federal Specification L–P–406, as follows:

Water absorption _____ method 7031
Heat distortion temperature _____ method 2011.1

4.4.1.2.1 *Deterioration tests.*

4.4.1.2.1.1 *Autoclaving.*—Plastic specimens and caps shall be autoclaved at 250°F. (15-17 pounds per square inch (p.s.i.)) for ten 30-minute periods, with a 10-minute rest period between each autoclaving.

4.4.1.2.1.2 *Chemical resistance.*—Samples of plastic cap material shall be tested as specified in method 7011 (salt solution immersion) of Federal Specification L–P–406.

4.4.2 *End item.*

4.4.2.1 *Bottle.*

4.4.2.1.1 *Annealing.*—Bottle shall be viewed in a polariscope to determine compliance with requirements for annealing, as specified in method 420 of Federal Test Method Standard No. 626.

4.4.2.1.2 *Bursting pressure.*—Each sample shall be completely abraded on its external surface, using 150-grit abrasive. External hydrostatic pressure shall be applied at a rate of 5 pounds per second to the sealed sample. The test shall be conducted at room temperature.

4.4.2.1.3 *Capacity.*—Capacity, using distilled water, shall be determined accurately to the nearest cc.

4.4.2.2 *Cap.*—Cap shall be autoclaved as specified in 4.4.1.2.1.1.

4.4.2.3 *Collar.*—The hardness of the collar shall be tested in accordance with method 3021 of Federal Test Method Standard No. 601. The rubber shall be tested unaged and

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after aging for 10 cycles of 30-minute duration in an autoclave at 250° F. (15-20 p.s.i.), with a 10-minute rest period between each autoclaving.

5. PREPARATION FOR DELIVERY

(For civil agency procurement Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.)

5.1 Packaging.

5.1.1 *Unit of issue.*—Six bottles, as specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE I.—*Packaging quantities*

Unit package	*Intermediate package*	*Exterior container*
1 unit	Not required	Level A – 4 units Level B – 1 unit

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit (6 bottles) shall be packaged in a regular, slotted, single-wall, corrugated, fiberboard box of not less than 200-pound Mullen test, with "A" or "C" flute corrugations, meeting all requirements of Federal Specification PPP–B–636. The interior of each box shall be fitted with full-size top and bottom pads and three full-height scored shells. The shells shall be scored and folded to form two individual cells or one double shell. All corrugations shall be horizontal. Shells shall provide two thicknesses of combined fiberboard between adjacent glass containers, and one thickness between glass containers and the inside faces of the ends and sides of the box. Scored sheets and pads shall be fabricated of the same material as the box. Bottles shall fit snugly in the cells. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be a box constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*—Unit package, as specified in 5.1.3.1, shall be shipped without further packing.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial

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containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civilian agencies.*

5.3.1.1 *Interior and exterior containers.* — In addition to any special marking specified in the contract or order, containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Interior packages.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2 (c), 6.2 (d), 6.5 and 7.5.1 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Intended use.**—Bottle is intended for use with Federal stock number 6530–702–9375, CABINET, SOLUTION WARMING, Two Section, Corrosion-Resisting Steel, 110 volt, 60 cycle, AC.

6.2 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Bottles, NOIBN other than cut, one quart or less in capacity.

Carload minimum weight 30,000 pounds per Uniform Freight Classification.

Truckload minimum weight 30,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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